Neural Computing and Applications (2025) 37:6995–7013
https://doi.org/10.1007/s00521-024-10456-z

REVIEW

Decoding silent speech: a machine learning perspective on data, methods, and frameworks

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Received: 18 February 2024 / Accepted: 9 September 2024 / Published online: 20 February 2025

Abstract

At the nexus of signal processing and machine learning (ML), silent speech recognition (SSR) has evolved as a game-changing technology that allows for communication without audible voice. This study offers a thorough overview of SSR, tracing its evolution from early waveform analysis to the most recent ML methods. We start by examining current SSR techniques using ML and determining the essential conditions for efficient SSR systems. After that, we look at the datasets and data collection techniques currently employed in SSR research, highlighting the difficulties posed by the variety of articulatory movements and the scarcity of data. Examining state-of-the-art SSR frameworks, the paper covers important topics such signal processing, feature extraction, ML techniques for decoding and optimizing and assessing the performance of SSR models. We emphasize how deep learning (DL) and ML models have evolved to increase SSR resilience and accuracy. The field's proposed procedures are examined, with an emphasis on sophisticated feature extraction and classification methods. Modern SSR techniques are compared in terms of performance, highlighting the advantages and disadvantages of different models. There is also discussion of ethical issues, especially those pertaining to privacy and consent. The integration of multimodal information—visual cues, electromyography signals, and neuroimaging data—to improve SSR systems is covered in this work. We investigate the functions of transfer learning and domain adaptation in handling cross-subject variability. Lastly, the study offers suggestions and future prospects for SSR research, providing practitioners, engineers, and academics with a road map. As SSR continues to push the frontiers of human–machine interaction, our study aims to increase our collective understanding of the technological advances and societal effects of SSR in the ML age.

Keywords Speech recognition · Silent speech · Machine learning · Deep learning · Speech decoding · Waves to words

1 Introduction

The confluence of machine learning (ML) and signal processing has made innovative solutions in human computer interaction (HCI) possible. Through the interpretation of articulatory gestures and delicate physiological indicators, silent speech recognition (SSR) is a paradigm-shifting discovery that enables communication without audible voice [34]. By deciphering articulatory and neuromuscular signals related to speech production, SSR offers a discreet, inclusive communication channel for people with speech problems [6]. SSR is still underreported in the literature, despite the focus on traditional spoken language recognition [14]. With a comprehensive analysis of SSR's worldwide reach and useful applications inside ML frameworks, this paper seeks to close this gap. SSR technologies have the potential to drastically change communication, particularly when ML and human–computer interaction converge [29].

As shown by Green et al. [19] using a Hidden Markov Model, single-word recognition is the main focus of current ML approaches for speech diagnostics in dysarthria or dysphonia. The differences between normal and damaged speech qualities make it difficult to develop automated speech recognition systems for individuals with speech disorders [41]. Our work treats this as a full-sentence

Extended author information available on the last page of the article categorization job, allowing patients to speak in a more natural way. This study highlights SSR's potential for accessibility and inclusion by following its evolution from early waveform analysis to sophisticated ML techniques [44]. Applications include hands-free communication and augmentative and alternative communication (AAC). We address issues like inconsistent articulatory motions and scarce data availability by analyzing SSR benchmarks and datasets [45]. To address these issues, further research avenues are recommended. Furthermore, we investigate how multimodal information may be used to improve SSR systems, including visual cues, neuroimaging data, and electromyography (EMG) signals. In order to overcome cross-subject variability, domain adaptation and transfer learning are investigated [14]. The purpose of this publication is to serve as a thorough resource for practitioners and researchers in the subject.

In order to bridge the gap between conventional single-word recognition and the requirement for natural, continuous speech recognition for people with speech impairments, this research represents some unique full-sentence classification strategy for SSR. In contrast to current approaches that only target single words, the systems explained in our study can identify full phrases too, facilitating more natural and fluid conversation. This is accomplished by utilizing cutting-edge ML methods, such as deep learning (DL) and neural networks, which can comprehend the intricacies of continuous speech patterns. The technologies greatly improve human–computer connection by deciphering the minute articulatory motions and brain signals involved with silent speech.

This paper is interesting since it provides a thorough overview of SSR research, highlighting the move toward full-sentence categorization and the use of multimodal data to improve system robustness and accuracy. It also discusses ethical issues and offers a clear plan for future developments in SSR technology research. This development facilitates more intuitive and natural interactions by offering more inclusive and effective communication options for those with speech problems. The objectives of this research are to: (1) summarize current SSR frameworks and methods; (2) assess important datasets and data collection procedures; (3) talk about integrating multimodal data for improved accuracy; (4) talk about consent and privacy ethics; and (5) offer a thorough road map for further SSR research. By means of these innovative contributions, this study hopes to direct future SSR research and development, fostering more efficient and inclusive communication tools.

The state-of-the-art in SSR research is covered in the sections that follow. "Related Works" offers a critical analysis of the literature, emphasizing approaches that further the discipline. We review key SSR datasets and data gathering techniques, providing an in-depth analysis of frameworks, structures, and features suggested by different research. The evolution of ML, and in particular DL models, is the main emphasis, as evidenced by the examined literature. While the "Limitations" section highlights challenge and acknowledges successes found in the evaluated research, the "Results" section provides empirical insights into model efficacy across datasets. The last section of the article, "Future Directions," offers a thorough summary of the state of the field now and its promise for the future by highlighting novel research avenues that have been brought to light by the literature review. Our study creatively synthesizes pioneering findings to trace the terrain of silent speech recognition through the perspective of ML, revealing paths from traditional single-word identification to more fluid, natural speech understanding. This analysis reveals the complex relationship between ongoing difficulties and technical advancement, encouraging more research in this revolutionary area.

2 Materials and methods

The methodologies employed in this study aim to advance the field of silent speech recognition (SSR) by leveraging cutting-edge ML and DL techniques. Given the interdisciplinary nature of SSR, our approach integrates sophisticated signal processing, feature extraction, and classification methods to decode silent speech with high accuracy and efficiency.

To ensure comprehensive coverage and robustness in our analysis, we conducted an extensive literature review and empirical studies, incorporating diverse data modalities and advanced ML algorithms. This section outlines the materials and methods used in our research, detailing the data collection processes, preprocessing techniques, feature extraction methods, and classification strategies. Additionally, we describe our search strategy and study selection criteria, ensuring a rigorous and systematic approach to compiling relevant literature and empirical evidence.

2.1 Search strategy

To ensure a comprehensive review of the methodologies used in silent speech recognition, we conducted an extensive search of relevant literature across multiple databases. This section outlines the search strategy employed to identify pertinent studies and references.

References for this review were identified through comprehensive searches of several databases, including MEDLINE/PubMed, Google Scholar, bioRxiv, arXiv, and IEEE Xplore Digital Library. The search strategy involved using a combination of ML keywords, speech-related keywords, and signal keywords. Specifically, the keywords used included:

ML Keywords: “deep learning”, “artificial intelligence”, “artificial learning”, “machine learning”, “neural networks”.

Speech-Related Keywords: “silent speech”, “speech recognition”, “decoding”.

Signal Keywords: “Speech Signal”, “Electroencephalography (EEG)”, “EEG Maturational Age (EMA)”, “Surface Electromyography (sEMG)”.

The search covered articles published up to January 2023. Additionally, relevant references cited in the identified articles were included to ensure comprehensive coverage of the topic.

2.2 Study selection

We applied specific criteria to select studies that provide valuable insights into decoding silent speech. This section details the inclusion and exclusion criteria used to identify relevant research for this review. The selection criteria for studies included in this review were as follows:

Inclusion Criteria:

- Articles in English that reported original research on decoding silent speech.
- Types of studies: development reports, implementation studies, clinical trials, or qualitative studies.
- Studies that provided proper documentation of experiments and quantitative results.

Exclusion Criteria:

- Studies that did not provide proper documentation of experiments.
- Studies that did not provide quantitative results.

In summary, our comprehensive approach involved a thorough search and selection process, advanced signal processing techniques, and state-of-the-art machine learning methodologies to decode silent speech. A total of 25 papers were meticulously selected for this review, providing a robust foundation for our analysis. By integrating multiple data modalities and employing sophisticated feature extraction and classification methods, we aim to enhance the accuracy and effectiveness of silent speech recognition systems. This methodological framework sets the stage for further advancements in the field, paving the way for practical applications in various domains.

3 Existing methods on silent speech recognition using ML

In order to identify the speech utterance, Silence Speech Interfaces (SSI) use inaudible data such as vibration, EMG, lip and facial pictures, Brain-Computer-Interface (BCI), ultrasound imaging, and WiFi signals [14]. Sun et al. [43] recorded the user’s lip movements to create a camera-based silent speaking interface that could recognize orders that were spoken. With SilentVoice, a microphone is positioned against the user’s mouth to capture “ingressive speech.” This method identifies the airflow during the whisper-like voice in order to establish silent conversation interaction. Moreover, The SottoVoce, a quiet voice interface based on ultrasound imaging and featuring a customized ultrasonic probe positioned beneath the user’s mouth, was created by Kimura et al. [29]. For SSI purposes, Kim et al. [27] used a magnetic tracer that was fastened to the tongue’s blade to record the movement of the tongue. Maruri et al. [35] introduced V-speech, which used a bone-conduction sensor found in the nasal pads of smart glasses to record voice signals.

The majority of active training and testing for SSI uses audible speech [45]. Although certain self-sensing interfaces rely on non-invasive technologies like ultrasonography or sensors placed on or near the lips and tongue, these interfaces were designed for healthy individuals and are therefore not yet suitable for use in actual clinical settings [2, 27, 44]. Denby et al. [12] demonstrated the potential advantages of further developing EMG-based silent speech interfaces, which still required mouthed words or facial motions, but were inexpensive, non-invasive, and suitable for usage with speech disorders like patients who underwent laryngectomy.

Voiceless silent speaking with minimal actions is a specialty of AlterEgo. This can be used in healthcare use cases, such as assisting patients with multiple sclerosis (MS) in communication, because the device is able to identify voiceless silent speech with minimum movements [26]. According to Cohen et al. [10], patients with MS currently use AAC equipment, such as dynamic touch displays, voice amplifiers, eye and head tracking technology, or communication boards with letters and symbols. Even if research has demonstrated that these communication techniques enhance quality of life in the early stages of neurodegenerative illnesses, there are a number of shortcomings. Gao et al. [14] presents “EchoWhisper,” a unique smartphone-based silent speaking interface that uses near-ultrasound waves to translate user speech without the need for any extra wearable sensors as shown in Fig. 1. Three benefits set EchoWhisper apart from other wearable and

Fig. 1 Visual representation of the benefits of EchoWhisper

mobile SSR technologies (such as lip reading, throat vibration, and airflow) in practical application scenarios.

4 Requirements of silent speech recognition using ML

The promise for revolutionizing human-device interaction lies in the creative combination of machine learning and signal processing, that is silent voice recognition. There are a few requirements that must be met in order for ML to effectively recognize silent speech. First, the quantity and quality of training data are fundamental; for machine learning algorithms to generalize well, a varied dataset capturing different articulatory patterns is required. For applications like assistive technology, real-time processing capabilities, feature extraction techniques, and an emphasis on articulatory subtleties are essential. Robust silent voice recognition systems must have both noise robustness in real-world contexts and adaptability and generalization over a wide range of user demographics.

Because the biometric data in question is sensitive, it is equally important to address user privacy issues and ethical considerations. Reliable silent voice recognition systems are built on a foundation of synergy between data quality, feature extraction, real-time processing, flexibility, noise robustness, and ethical concerns. This all-encompassing strategy guarantees that the creative potential of the technology is utilized in a variety of applications. Enabling the full potential of machine learning for silent voice recognition still depends on satisfying these prerequisites as academics and practitioners explore this revolutionary topic [21].

5 Current data acquisition methods

Careful data gathering techniques are essential to the development of ML and DL models for silent voice recognition. Silent speech is secretive and internal, which presents unique difficulties. This emphasizes the requirement for methodological accuracy in gathering pertinent data for the models' training and evaluation stages. A representation of such data acquisition procedure is illustrated in Fig. 2. Traditional strategies, such as relying on eye movements, were initially unsuccessful in Chaudhary et al. [7]. The neurofeedback-based paradigm, introduced on the 86th day after implantation, involved providing auditory feedback based on neural activity, enabling the patient to modulate neural firing rates successfully. Data collection spanned from day 106 to day 462 post-implantation, with multiple neurofeedback sessions conducted during this period.

The Silent Speech Challenge dataset (SSC) data acquisition system in Yan et al. [49] involves a helmet with a 128-element ultrasound probe and an infrared-illuminated video camera capturing tongue and lip images at 60 frames per second. The training corpus includes ultrasound and lip video data from a native English speaker, while the test set comprises sentences from the WSJ0 5000-word corpus. The archive contains raw ultrasonography and lip images, along with original and reduced-length feature vectors. The signals of Kapur et al. [26] are collected non-invasively from eight electrodes attached to the user's face and neck regions. The dataset consists of recordings from three MS patients, each articulating 15 different sentences 10 times. In Meltzner et al. [36], the sensors were placed over specific muscles associated with speech production. The study utilized three different corpora (Corpus 1, Corpus 2, and Corpus 3) for experimentation. Corpus 1 involved isolated words, Corpus 2 consisted of small vocabulary sequences of words, and Corpus 3 included a large vocabulary of continuous speech.

Ravenscroft et al. [38] focused on creating an automated method for SSR using a graphene strain gage sensor. The data acquisition process involved three main steps: Device Fabrication, Data Generation, and ML Classification. A graphene strain gage sensor was fabricated to detect signals related to speech, worn on the throat. Data were generated by collecting a dataset comprising repetitions of 15 words and movements, recorded using the strain gage sensor. The data used in Cai et al. [5] were derived from 50 healthy participants, aged between 20 and 30, whose first language is Mandarin. The dataset contained one blank instruction

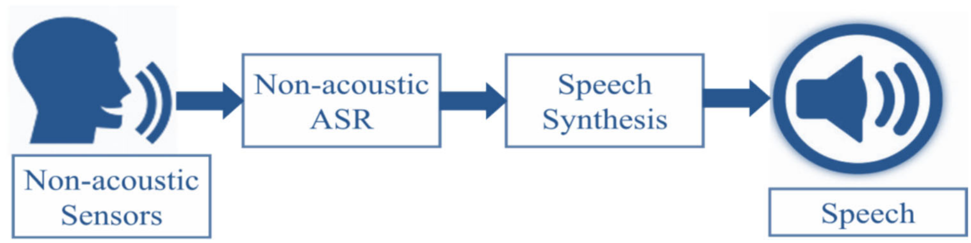


Fig. 2 Data acquisition for silent speech recognition. ASR; Automatic Speech Recognition

and 100 distinct phrase instructions. Every instruction had a duration of two seconds. All of the participants in Cha et al. [6] experiments conducted using written and audio instructions while seated in an easy chair in front of a liquid crystal display computer monitor. E-prime 2.0 (Psychology Software Tools, Sharpsburg, PA, USA) was used to provide the instructions. It also captured the precise moments when the participants gave a silent speech. The multichannel EMG amplifier EMG-USB2, manufactured and distributed by OT Bioelettronica, Italy was utilized for EMG recording in Wand et al. [45]. With a recording bandwidth of 3 Hz–4400 Hz and a customizable strength of 100–10,000 V/V, the EMG-USB2 amplifier can record and process up to 256 EMG channels. The participants were instructed to envision speaking the labeled words displayed on a computer screen one by one in a predetermined sequence, which is the meaning of silent speech, while the six-channel sEMG is recorded in Wang et al. [46]. Ten Chinese words and a recording bandwidth of 3 Hz–4400 Hz were used in the tests.

6 Dataset used in this domain

The dataset of Chaudhary et al. [7] encompassed neural activity recorded from the implanted microelectrode arrays during communication sessions. Neurofeedback trials, both with and without reward, were conducted over 281 blocks, comprising a total of 1176 sessions. The study utilized data from the patient's communication attempts, including neurofeedback blocks directly preceding speller sessions, involving both successful and unsuccessful trials. This publication has source data attached. The raw brain recordings are accessible to the author upon request, however due to the potentially sensitive nature of the data, sharing these datasets will need an agreement between the Wyss Center and the researcher's university.

The SSC training corpus of Yan et al. [49] includes ultrasonography and lip video data from a native English speaker pronouncing TIMIT corpus utterances. The test set comprises sentences from the WSJ0 5000-word corpus. The data are publicly available for research purposes [11]. The study Kapur et al. [26] recorded a dataset of silent speech from three MS patients using AlterEgo, a peripheral neuromuscular-computer interface. The signals of interest were neuromuscular recordings resulting from disruptions in the electric field around internal speech articulators and surface electromyography (sEMG) signals. The SSC dataset comprised tongue and lip images captured at 60 fps, with tongue images having dimensions of 320×240 pixels, and lip images having dimensions of 640×480 pixels. The dataset is not publicly available.

The training set of Kimura et al. [29] consisted of 2342 utterances from a single native English speaker reading the TIMIT corpus without vocalization. This was not found publicly. The study Meltzner et al. [36] involved recordings of sEMG signals from articulator muscles of the face and neck during subvocal speech tasks with some publicly available datasets [13, 15, 37].

A total of 19 subjects participated, and sEMG signals were acquired using both tethered DE 2.1 sensors and prototype wireless TrignoTM Mini sensors. Two distinct datasets are developed in Ravenscroft et al. [38]—one for words and another for noises/head movements. The word dataset includes 15 phonetically balanced words with 20 repetitions each. The movements dataset consists of recordings of coughing, swallowing, yawning, and nodding. The dataset was designed to be suitable for training classification algorithms, emphasizing the importance of repetitions to discern features consistent across the same word or movement. There were ten words and ten phrases in the open-source dataset Miracl-VC1 that are specifically intended for lipreading which is used in the study of G. Kumar and William [32]. 15 persons uttered each word and phrase in the dataset ten times, for a total of 3000 utterances. Ten words in all (begin, choose, hello, web, start, next, stop, connection, navigation, and previous) can be found in the Miracl-VC1 dataset. The dataset is available on request from the corresponding author.

The ZuCo datasets used by Wang et al. [47] comprised simultaneous EEG and eye-tracking data obtained from real-world reading activities which is available upon request from the authors. In Cai et al. [5], face EMG signals were collected using a six-channel EMG device with a 1 kHz sampling frequency. The electrodes were positioned in the following facial muscles: mylohyoid, platysma, levator labii superioris, anterior belly of the stomach, and mentalis. The data (publicly not available) was derived from fifty healthy individuals, aged between twenty and thirty, whose first language is Mandarin. Hahm et al. [20] experiment made use of the publicly available MOCHA (Multi-CHannel Articulatory)-TIMIT Data of University of Edinburgh [1]. The 920 sentences in the MOCHA-TIMIT dataset were taken from the TIMIT database which has open access. This consists of the contribution of two British English speakers: one female (FSEW0) and one man (MSAK0). This dataset included concurrent speech recordings, information on articulatory movements, and more data types. Tóth et al. [44] involved BEA Hungarian Spoken Language Database [18] with 84 individuals in total. Out of this, 36 were healthy controls and 48 were mild cognitive impairment (MCI) patients. The 48 MCI individuals were all native Hungarian speakers who spoke the language right-handed and had no prior medical history of hearing loss. The datasets are publicly available. Among the various methods utilized by authors, surface electromyography (sEMG) is the most prevalent. Figure 3 presents a graphical representation of the different speech data modalities and the quantities used in the reviewed studies.

7 Current frameworks

In the realm of silent speech recognition, the translation of non-audible vocal gestures into discernible language via machine learning presents a unique set of mathematical and computational challenges. This section delves into the mathematical frameworks that underpin the entire process—from signal acquisition to the decoding of these signals into verbal communication. Meticulous formulation is crucial, as it directly influences the accuracy and efficiency of the recognition systems.

We begin by detailing the methodologies employed in the capture and preprocessing of signals, which are vital for ensuring the quality and usability of the data input into machine learning algorithms. Subsequent discussions focus on the extraction of relevant features from these processed signals using robust mathematical models. This is followed by an exploration of various machine learning algorithms, each defined by specific mathematical structures, which facilitate the mapping of extracted features to speech output. Finally, we examine the optimization strategies and performance metrics that are essential for refining system performance and ensuring practical viability in real-world applications.

7.1 Signal processing and feature extraction

The mathematical representations of silent speech signals have been the subject of recent research. Important developments concern the application of mathematical transforms, including Wavelet Transform and Short-Time Fourier Transform (STFT), to capture the spectral and temporal subtleties present in hidden articulations. Improving signal representation has centered on creating feature vectors that capture crucial data from accelerometry, EMG, and other physiological signals.

Meltzner et al. [36] utilized a linear discriminant analysis (LDA) algorithm [39] to explore various numbers of

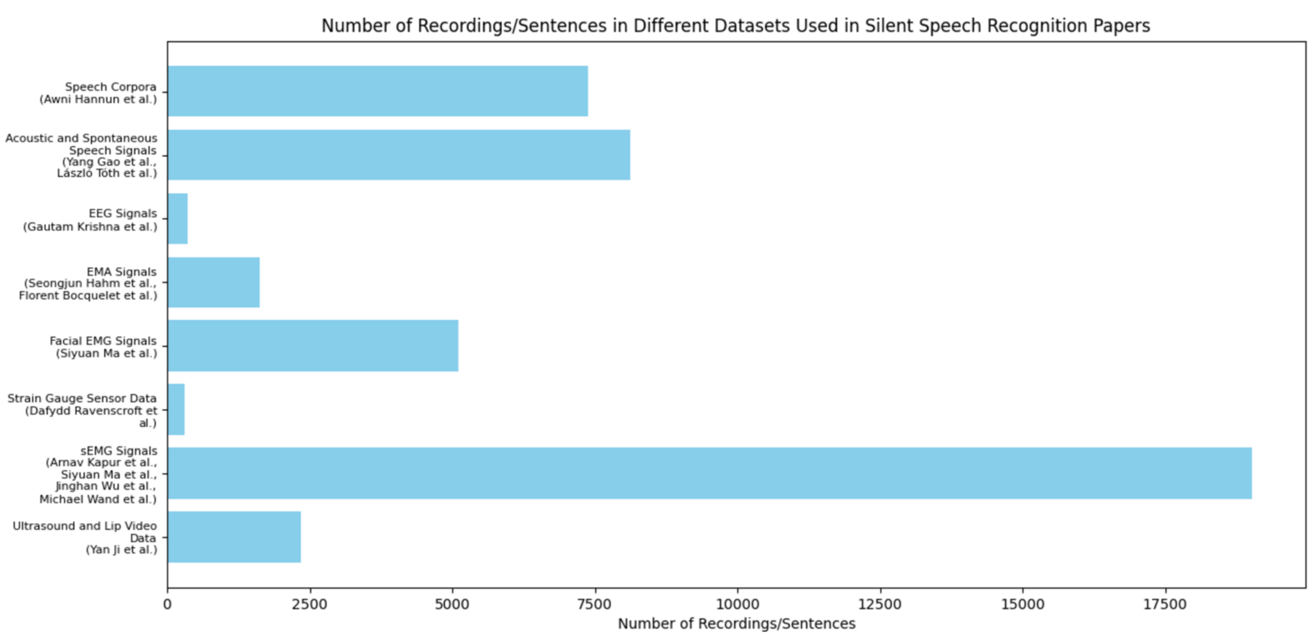


Fig. 3 Quantitative comparison of different speech data modalities used in silent speech recognition studies

reduced features by transforming 154 Mel-Frequency Cepstral Coefficient (MFCC) features from 11 sEMG channels into a lower-dimensional space. Feature reduction is crucial in pattern recognition to enhance classification accuracy, decrease processing times, and reduce memory needs. Previous research has applied variations of LDA to silent speech recognition (SSR) with simpler vocabularies [50]. We used a heteroscedastic LDA (HLDA), which unlike standard LDA, does not assume uniform within-class covariance matrices. This HLDA approach allowed us to efficiently map a high-dimensional space to a smaller one, optimizing via a maximum likelihood function [33].

$$Q\left(\theta,\hat{\theta}\right)=\sum_{m\epsilon M}\gamma_m(\tau)\log\left(\frac{\hat{A}^2}{\left|diag\left(\hat{A}_pW^{(m)}\hat{A}^T_{p)}\right)\right|\left|diag(\hat{A}_{n-p}T\hat{A}^T_{n-p)})\right|}\right)$$

where θ stands for the older model, $\widehat{\theta}$ stands for the newer model, $\widehat{A}$ is the inverse transformation matrix. $\widehat{A}_p$ and $\widehat{A}_{n-p}$ are the corresponding first p and remaining $n-p$ rows.

$$\gamma_m(\tau)=p(q_m(\tau)|\theta,O_T)$$

$$W^{(m)}=\frac{\sum_\tau\gamma_m(\tau).(o(\tau)-\widehat{\mu}^{(m)})(o(\tau)-\widehat{\mu}^{(m)})^T}{\sum_\tau\gamma_m(\tau)}$$

$$\widehat{\mu}^{(m)}=\frac{\sum_\tau\gamma_m(\tau).o(\tau)^T}{\sum_\tau\gamma_m(\tau)}$$

$$T=\frac{1}{T}\sum_\tau(o(\tau)-\widehat{\mu}^{(g)}).(o(\tau)-\widehat{\mu}^{(g)})^T$$

Here, $\widehat{\mu}^{(g)}$ represents the overall mean of the data, and $\gamma_m(\tau)$ denotes the probability that the Gaussian component m at time τ matches the previous model and the observed feature sequence $o(\tau)$. The authors refined the cost function using a conventional expectation maximization (EM) algorithm, setting a cap on the number of iterations to prevent overfitting.

7.2 Machine learning algorithms for decoding

In the pursuit of translating silent speech into intelligible language, the application of machine learning algorithms plays a pivotal role. This section of the paper focuses on the diverse array of algorithms specifically tailored for decoding the complex patterns inherent in silent speech recognition. The mathematical underpinnings of neural network architectures are essential to ML. Sophisticated recurrent neural networks (RNNs), convolutional neural networks (CNNs), and attention mechanisms have evolved from traditional architectures, as demonstrated by recent publications. These architectures are painstakingly designed to extract complex patterns from silent speech signals, allowing for more precise and context-aware detection [31].

Hannun et al. [21] employed a recurrent neural network (RNN) model featuring five layers of hidden units. The notation $h^{(l)}$ is used to denote the hidden units in layer l, where $h^{(0)}$ represents the input layer. The first three layers of this model are non-recurrent. Specifically, in the first layer, the output at each time t is influenced by the spectrogram frame x_t and includes a context of c frames from each side. The subsequent two non-recurrent layers process data independently at every time step. Therefore, the computations for the first three layers at each time t are as follows:

$$h_t^{(l)}=g(W^{(l)}h_t^{(l-1)}+b^{(l)})$$

In this model, (g(z) = min (max (0, z), 20)) serves as the clipped rectified-linear (ReLU) activation function, while $W^{(l)}$ and $b^{(l)}$ represent the weight matrix and bias parameters for layer l, respectively. The fourth layer is configured as a bi-directional recurrent layer [40].

Using the output $\mathbb{P}(c|x)$ from the RNN, the authors conducted a search to identify the sequence of characters $c_1,c_2,c_3,\ldots\ldots$ that is deemed most probable based on both the RNN's output and the language model, which interprets the sequence of characters as words. Specifically, the goal here was to identify a sequence c that maximizes the combined objective:

$$Q(c)=\log(\mathbb{P}(c|x))+\alpha\log(\mathbb{P}_{lm}(c))+\beta\text{word\_count}(c)$$

Here, α and β are tunable parameters, determined through cross-validation, that manage the balance between the RNN output, the constraints imposed by the language model, and the length of the sentence. The term $\mathbb{P}_{lm}$ represents the probability of the sequence c as computed by the N-gram model.

Ravenscroft et al. [38] addressed the supervised classification challenge of mapping extracted features to their associated noise using machine learning models, specifically random forests (RFs) and k-nearest neighbors (k-NN). RFs utilize a combination of decision trees to form a robust classifier, while k-NN, a non-parametric method, predicts outcomes based on the nearest training samples [4]. The authors designed a neural network featuring a convolutional layer paired with Long short-term memory (LSTM) layers, as illustrated in Fig. 4. This setup enabled the identification of diverse temporal features. To mitigate overfitting, dropout layers and a ReLU layer are integrated into the network. A softmax layer functions as the classifier, facilitating the computation of error throughout the network. In order to assess the algorithm's practical applicability, the suggested neural network model's running duration was assessed. The wearable graphene strain

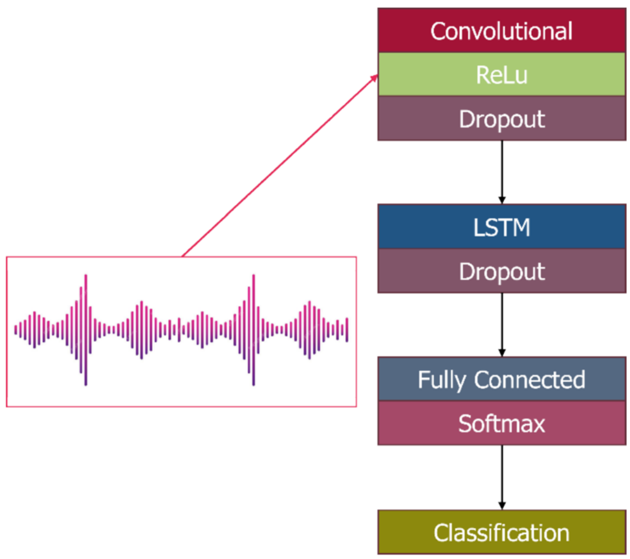


Fig. 4 The neural network developed for signal to phonation classification incorporates a convolutional layer followed by a ReLu Activation Layer, an LSTM Layer, and Two Dropout Layers to Prevent Overfitting. Additionally, It Includes a Fully Connected Layer and Employs a Softmax Classifier to Categorize the Outputs

gage sensor captures accurate neck movements and enhances automatic quiet speech identification due to its exceptional wearability and flexibility. Its quick and simple manufacturing process increases the efficiency of its deployment. Because graphene is a strong, flexible, and highly conductive material, machine learning algorithms can accurately and efficiently predict speech. By taking advantage of the neural network's sophisticated feature extraction capabilities and its ability to accurately capture neck movements, integrating a sensor of this kind with the neural network model presented by Ravenscroft et al. might improve the detection of silent speech. About 100 ms to 1.5 s would be a fair estimate of how long the algorithm would take to process and categorize the signals from the graphene strain gage sensor. This range covers the preparation, data collection, and machine learning algorithm execution on standard computing hardware.

On the other hand, the neural network model presented by Cai et al. [5] achieved an average inference time of 0.0167 s per sample on the same representative dataset, which took 0.2 s using sample entropy algorithm method. The model demonstrates its suitability for real-world silent speech recognition tasks using facial electromyography. It presents an end-to-end deep neural network model for speech recognition, building on previous frameworks. Initially applied by Jiang et al. for multi-label image classification, this model combines a CNN with an RNN structure [25]. This research further refined this approach using pre-trained MobileNet-V1, ResNet18, and Xception models to extract features from spectrograms, followed by Bi-LSTM and GRU classifiers for prediction [47]. Our study advances this model by simplifying the CNN network at the front end to achieve comparable performance. This model utilizes CNN for initial feature extraction and a Bi-GRU for decoding at the back end. The network culminates in a fully connected layer outputting 101-dimensional data, with classification results derived using the SoftMax function.

7.3 Optimization and performance evaluation

Model optimization has benefited greatly from developments in the mathematical formulation of training objectives. The focus of recent research has been on new loss functions designed specifically for SSR that include weighted penalties for incorrect classifications resulting from minute articulatory motions. Regularization methods used in conjunction with adversarial training approaches help to increase the generalization and robustness of the model [7].

Hannun et al. [21] implemented several strategies to enhance the speed and efficiency of network training. They chose to use homogeneous rectified-linear networks for their simplicity and reliance on a minimal number of optimized BLAS operations. With nearly 5 billion connections unrolled per typical utterance, ensuring efficient computation was crucial. To facilitate this, the KenLM toolkit [22] was used to train the N-gram language models. Additionally, they employed multi-GPU training to further accelerate the process, although this approach required additional optimization steps to be effective [9].

Adding neurophysiological data—like EEG signals—to the mathematical framework adds a level of complexity. In order to efficiently fuse information from many modalities, researchers have created mathematical models. This entails creating learning objectives that span several modalities and include layers of neural networks that can process both physiological signals and conventional aspects of silent speech [6].

Recurrent Neural Networks (RNNs) excel in sequential data processing, especially in natural language applications. Unlike MLPs and CNNs, RNNs can use their outputs as inputs for subsequent layers, enabling information persistence [42]. However, they face challenges with long sequences due to the vanishing gradient problem during backpropagation [8].

Long Short-Term Memory (LSTM) networks, a type of RNN, address this by using gates (forget, input, output) to manage information flow, allowing them to maintain long-term dependencies effectively [17]. Extending this, Wang et al. [47] used a bidirectional LSTM (bLSTM) that utilizes both forward and backward processing to capture semantic dependencies from all directions, making it ideal for complex applications like six-channel sEMG. The authors also employ Multi-Layer Perceptron (MLP) and Convolutional Neural Network (CNN) in their work. Each decoder is designed with input layers, hidden layers, dense layers, and softmax output layers. The MLP utilizes fully connected layers with dropout for regularization and ReLU activation functions to enhance performance. CNN employs convolutional and pooling layers to capture spatial features, incorporating batch normalization to stabilize the learning process. These architectures are tailored to learn and predict silent speech from the sEMG features, with a comparative analysis to determine the most effective method.

The steps that a neural network carries through in order to decode spectrogram pictures and predict labels are shown in Fig. 5. The visual depiction of the frequency spectrum throughout time is first shown in spectrogram pictures. The Xception model processes these photos and extracts a feature set. The neural network is composed of six layers, each with 1000 units, and the input layer is made up of the retrieved characteristics. The neural network's hidden layers, which carry out further processing and feature abstraction, receive input from the input layer. After processing, the data is sent via a thick layer that is completely linked and merges the data from the hidden layers. The output is then processed by a softmax layer, which produces the anticipated labels from the models by converting the network output into probabilistic predictions.

Recent evaluations have highlighted the importance of the mathematical foundation for transfer learning and adaptability. Researchers investigate how to apply insights from related datasets of audible speech to customize models for specific users. To improve model performance in practical, user-specific circumstances, this entails creating adaptation strategies and transfer learning objectives [14, 44].

The progress of the mathematical formulation framework—which includes signal representation, neural network architectures, training objectives, integration of neurophysiological data, decoding mechanisms, and transfer learning strategies—is highlighted by recent efforts in SSR. A substantial advancement in the field of ML-driven silent speech detection has been made with the synthesis of these mathematical formulations, which show a concentrated attempt to bridge the gap between the complexities of silent speech and the capabilities of ML algorithms.

The importance of several optimization techniques in furthering SSR is highlighted in this section. Use of multimodal data fusion, weighted penalties, and sophisticated architectures like CNNs and bLSTMs show important advances in managing SSR complexity. We can increase scalability and performance by putting effective strategies

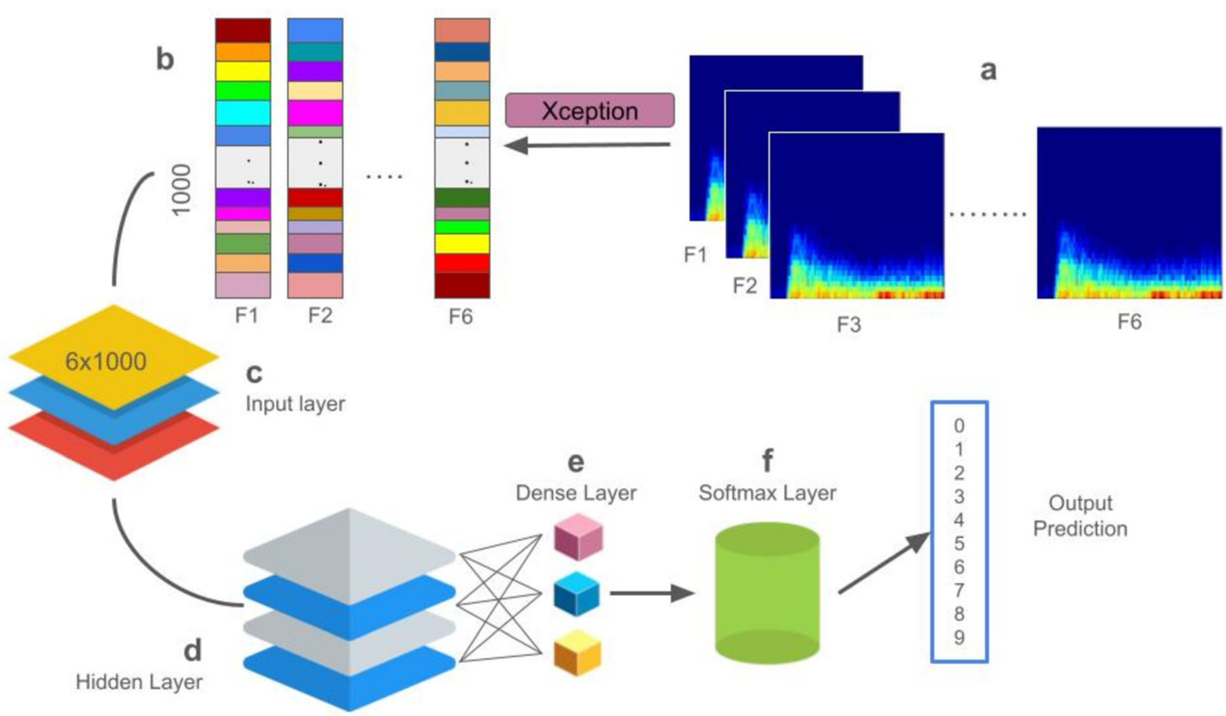


Fig. 5 Decoding processes: **a** Spectrogram Images, **b** Feature Set Extracted by Xception, **c** Input layer of Neural Networks, **d** Hidden Layers of Neural Networks, **e** Fully Connected Dense Layer, **f** Softmax Output Layer, and **g** Predicted Labels Obtained from the Models

like improved BLAS operations and multi-GPU training into practice. These techniques emphasize the critical trade-off between robustness of the model and computing efficiency. In addition to advancing SSR, the insights presented here also set the stage for future optimization approaches targeted at improving performance even further in this developing sector.

7.4 Current ML and DL models for silent speech recognition

The quest for increasingly efficient ML and DL frameworks is causing a revolution in the field of silent voice recognition. There is a noticeable change as we dive into the core of recent research—a progression in the mathematical expressions that support the creation of models that can understand the complexities of interior speech. A preview of the suggested ML and DL frameworks is given in Table 1; they each represent a patchwork of creativity meant to reveal the hidden possibilities of silent voice detection.

Figure 6 illustrates how various neural network topologies and techniques are used to transform EEG information into speech recognition. The method starts with the gathering of EEG data, which is subsequently processed through a number of layers, such as a Temporal Convolutional Network (TCN) with 64 units and a Gradient-Based Rule Utility (GRU) with 128 and 64 units. Connectionist Temporal Classification (CTC) loss is used to the processed data in order to match the predictions to the target sequences. The results are then classified into probabilistic distributions across potential characteristics by using a softmax layer. Lastly, silent speech is generated by using a time-distributed dense layer to anticipate the spoken text across time.

The exploration of proposed ML and DL models highlights the adaptable and creative methods used for silent speech interpretation. The techniques explained, which range from the complexities of neural network designs to the use of sophisticated training algorithms, represent a cooperative effort to improve the accuracy and performance of silent voice recognition systems. The amalgamation of computational methodologies and cognitive science insights not only reflects the multidisciplinary nature of this domain but also underscores the possibility of revolutionary progressions (Kim et al. [28]). Moving forward, these suggested models create a foundation for further investigation and improvement, adding a great deal to the always changing field of silent speech detection using ML and DL approaches. A silent voice detection system is depicted in Fig. 7, along with the steps involved in data collecting and identification. After gathering the raw data, it is preprocessed to make it clean and ready for analysis. The preprocessed data is then used to extract pertinent characteristics. Model testing and model training are the two key stages of the system. In the model training stage, labeled data is utilized to train a machine learning model using the extracted characteristics in order to comprehend and anticipate patterns of silent speech. In order to assess the model's performance in real-world circumstances, new data is subjected to the learnt parameters from training.

8 Proposed methodologies in this domain

In this section, we present a thorough methodology that is at the forefront of using cutting-edge ML and DL techniques to advance silent voice detection. Approach combines sophisticated signal processing, neural network topologies, and state-of-the-art training techniques, all carefully crafted to reveal the subtleties of interior speech. Together, all the parts work to create a strong framework as portrayed in Fig. 8 that improves efficiency and accuracy while also setting the stage for revolutionary developments in the field of silent voice detection.

8.1 Feature extraction

Recent advancements in feature extraction techniques have profoundly impacted the development of SSR systems, which are increasingly recognized as essential tools for individuals with speech impairments, communication in noisy environments, and hands-free device control. These techniques are crucial for capturing the subtle and complex nuances of silent articulations, which lack the overt acoustic cues of audible speech. This section reviews the literature, summarizing key methodologies and innovative approaches in feature representation that enhance the accuracy, robustness, and applicability of SSR systems across various domains.

1. **Temporal-Spectral Representations:** Capturing the dynamic nature of silent speech signals has been a focal point in SSR research, with temporal and spectral representations emerging as effective tools. The application of techniques such as the Wavelet Transform and STFT has shown promise in isolating and emphasizing the spectral properties and temporal fluctuations of silent articulations. These methods provide a robust framework for constructing feature vectors that capture the fine-grained details necessary for precise recognition. In practical applications, these representations are particularly valuable in scenarios where distinguishing between subtle articulatory variations is critical, such as in the development of speech prostheses and real-time silent communication systems [26].

Table 1 Proposed ML and DL Models for Silent Speech Recognition

Reference	Model	Approach
Ji et al. [24]	Deep Auto Encoder (DAE)	The encoder processes the input data, gradually reducing its dimensionality, and the decoder reconstructs the original input from this reduced representation. The process involved selecting Regions of Interest (ROIs) from tongue and lip images and resizing them, for computational tractability
[26]	Convolutional neural network (CNN)	The optimal number of convolutional layers and their configurations were determined through progressive design and tuning. After each convolutional layer, pooling layers are applied to reduce the spatial dimensions of the data, retaining essential features while decreasing computational complexity
[29]	End-to-End Speech Processing Toolkit, ESPnet	ESPnet is an open-source toolkit that facilitates end-to-end speech processing tasks. The employed ASR model is based on the VoxForge recipe of ESPnet, which uses a hybrid CTC / attention-based end-to-end ASR model as shown in Fig. 5
[36]	Random Forests (RF) and k-Nearest Neighbor (k-NN)	A Convolutional-Recurrent Neural Network (NN) is employed for automatic feature learning and classification. The neural network architecture includes a convolutional layer, a ReLU layer, an LSTM layer, dropout layers, and a softmax classifier
Bocquelet et al. [3]	Articulatory hidden Markov model (HMM) and acoustic HMM	The adaptation process was split into two stages: the real-time control testing stage, where the articulatory synthesis was carried out in real-time in accordance with the subject articulatory movements in the silent speech condition, and the calibration step, where an articulatory-to-articulatory mapping was estimated to map the EMA trajectories of the on-line session onto those of the reference off-line session
[16]	MobileNet, LSTM, Support vector machine (SVM), Histogram of oriented gradients (HOG)	To get weights for the corresponding labels, a linear SVM classifier is trained on positive and negative samples. The probability of classification and false-positives from the training set are then obtained using the hard-negative mining technique. In order to retrain the classifier, the false-positive samples that came from the hard negative mining are arranged based on the probabilities. For each image in the test set, the aforementioned technique is run again to produce HOG descriptors for classification
Wu et al. [48]	Parallel-Inception CNN (PICNN)	The input features of PICNN were analyzed in parallel according to their channels. To create a single inception module for each channel and carry out the convolution process concurrently, filters of various widths were used
[34]	SVM, k-NN, RF	Over training, each feature was given a score. The trait with the higher score was more significant. We employed a number of popular ML techniques for pattern recognition after getting the feature datasets
Kim et al. [28]	Hidden Marcov Model- Gaussian Mixture Model (GMM-HMM)	The number of states of the monophone and triphone GMM-HMM are represented by the 113 and 399 softmax output layers and the 256 hidden units at each layer of the DNN, respectively. Layer-by-layer generative pre-training based on restricted Boltzmann machines (RBM) was used to establish the parameters

2. **DL-Based Representations:** The integration of DL techniques into silent speech feature extraction marks a transformative shift in the field. Models such as CNNs and Recurrent Neural Networks RNNs have demonstrated the ability to autonomously learn hierarchical features from raw data. This capability allows for the extraction of intricate patterns from silent speech signals, thereby enhancing the model's understanding of the nuanced articulatory movements that characterize silent speech. These DL-based representations are increasingly being employed in advanced SSR systems designed for assistive technologies, such as devices enabling communication for individuals with conditions like amyotrophic lateral sclerosis (ALS) or stroke-induced aphasia, where traditional speech recognition methods fall short [21, 38].
3. **Physiological Signal Integration:** The integration of physiological signals, such as accelerometry and EMG,

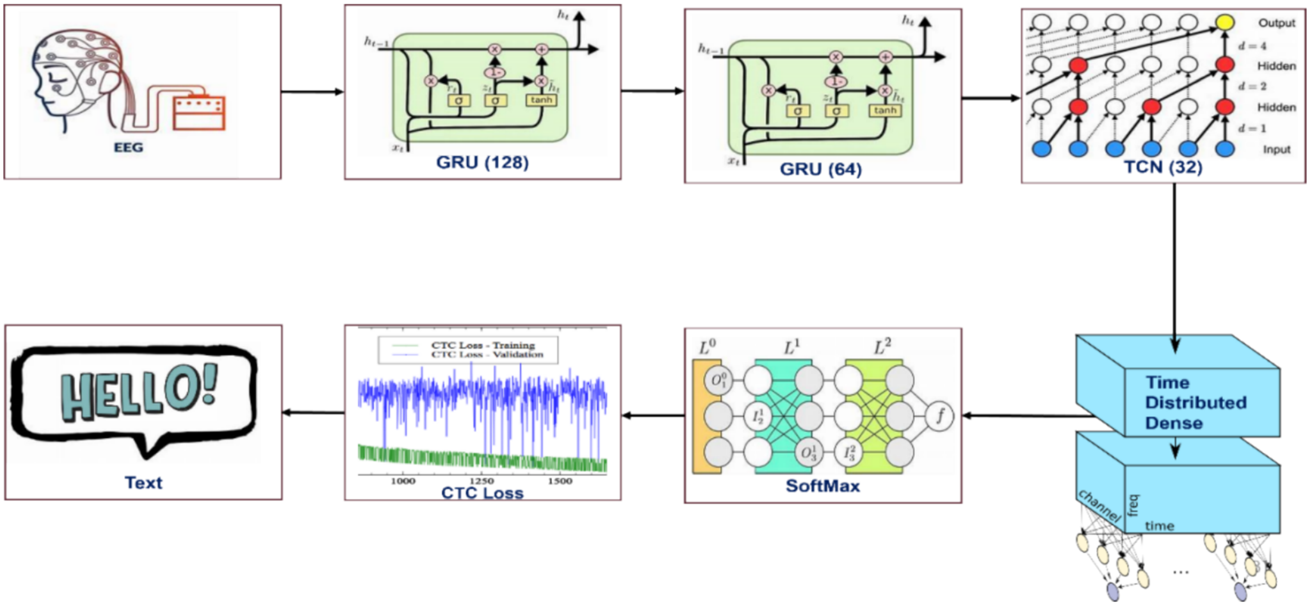
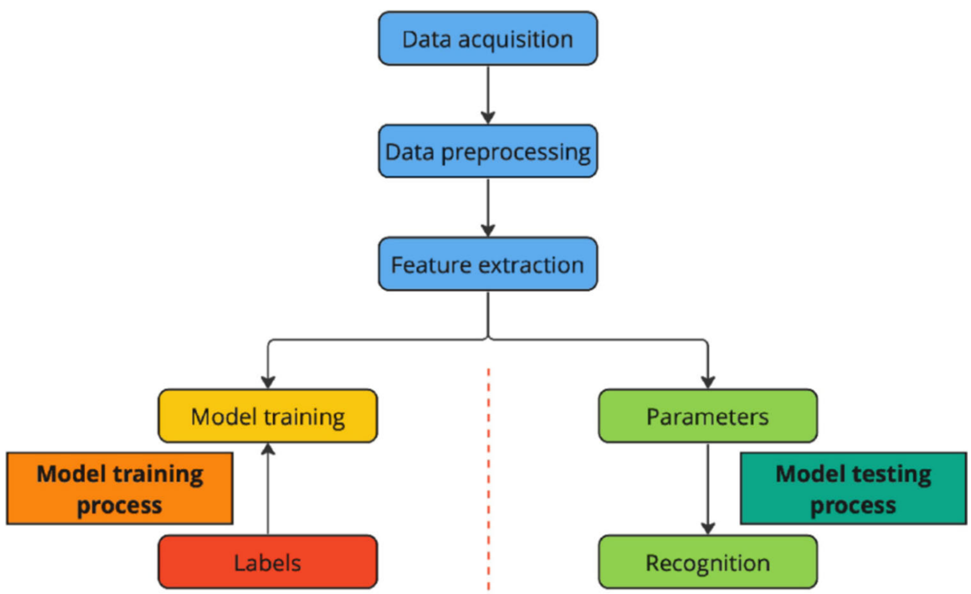


Fig. 6 CTC automatic speech recognition model. EEG, electroencephalogram; GRU, Gradient-Based Rule Utility; TCN, Temporal Convolutional Networks; CTC, Connectionist Temporal Classification

Fig. 7 Silent speech recognition system



into the feature extraction process represents a significant advancement in SSR technology. These multimodal approaches combine traditional audio data with physiological inputs, capturing complementary information that enriches the feature space and enhances recognition accuracy. This technique is particularly relevant in developing robust SSR systems for environments where traditional audio-based methods are ineffective, such as in high-noise settings or for users with severe speech impairments. The incorporation of physiological signals has been shown to improve the performance of SSR systems in real-time applications, including silent command and control interfaces for military or industrial use, where silent communication is paramount [5, 6].

4. **Frequency-Domain Features:** Frequency-domain features, especially MFCCs, continue to play a foundational role in SSR systems. Despite the evolution of feature extraction techniques, MFCCs remain integral due to their ability to capture the spectral characteristics essential for speech recognition. When combined with advanced methods such as Gammatone filter

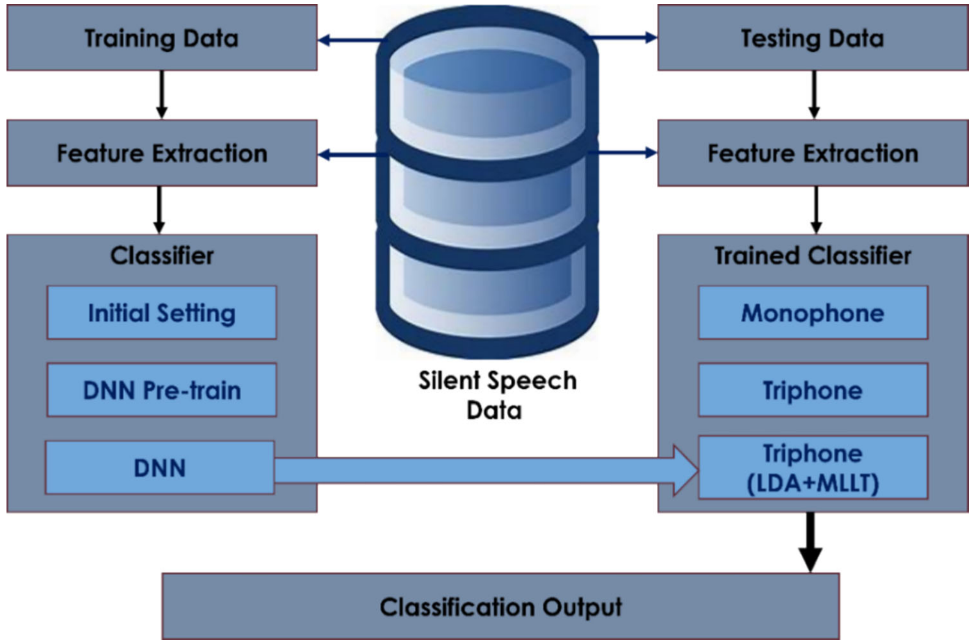


Fig. 8 Classification by feature recognition. LDA, Linear Discriminant Analysis; MLLT, Maximum Likelihood Linear Transform; DNN, Deep Neural Network

banks, these features provide enhanced spectral representation of silent speech signals, crucial for accurate recognition across various contexts. This approach is particularly useful in applications such as silent command systems for consumer electronics and wearable devices, where efficiency and reliability are paramount [30, 36].

5. **Context-Aware Representations:** The inclusion of contextual information in feature extraction has emerged as a critical factor in improving SSR performance. Context-aware representations, such as those derived from contextual embeddings and attention mechanisms, allow for the fine-tuning of feature vectors by considering the sequential dependencies and contextual cues present in silent speech. This approach significantly enhances the recognition accuracy of SSR models, making them more adaptable to naturalistic communication scenarios where context plays a vital role. These representations are particularly valuable in conversational AI systems and augmented reality (AR) applications, where understanding user intent in context-rich environments is crucial (Kim et al., [28]; [34]).
6. **Hybrid Approaches:** Hybrid feature extraction techniques, which combine traditional methods with DL, have gained traction as a means of leveraging the strengths of both approaches. By integrating manually engineered features, such as formant frequencies, with features learned through deep neural networks, these hybrid models aim to optimize feature extraction by combining domain-specific knowledge with the automatic learning capabilities of DL models. This hybridization is especially beneficial in the development of SSR systems for specialized applications, such as medical diagnostics and rehabilitation tools, where both precision and adaptability are essential (Bocquelet et al., [3]).

The practical applications of these advanced feature extraction techniques in SSR are already making significant impacts in various real-world scenarios. For instance, the integration of temporal-spectral representations and deep learning models has been instrumental in developing assistive communication devices for individuals with speech impairments, such as those affected by ALS or severe vocal cord damage. These devices allow users to communicate effectively by interpreting silent articulations and translating them into synthesized speech. In high-noise environments, such as in military operations or industrial settings, SSR systems employing physiological signal integration, including EMG and accelerometry, offer robust silent communication methods that bypass the limitations of traditional audio-based systems. Moreover, the application of context-aware and hybrid approaches has enhanced the functionality of wearable technology and smart devices, enabling silent command recognition for hands-free operation in everyday activities, such as controlling home automation systems or interacting with AR interfaces. These real-life implementations not only demonstrate the versatility of SSR technology but also highlight its growing importance in creating inclusive, efficient, and contextually aware communication solutions.

Novel developments in feature extraction techniques for silent voice identification demonstrate a wide range of strategies. The literature displays a dynamic landscape of innovation, ranging from the integration of physiological inputs and context-aware representations to temporal-spectral representations and DL-based features.

8.2 Classification techniques

Strong classifiers that can reliably identify patterns in feature-rich representations as shown in Fig. 5 are essential to the advancement of silent voice detection. In this section, recent research is reviewed with an emphasis on the wide range of classification approaches used to convert complex feature sets into precise and context-aware predictions in the field of silent voice detection.

1. **DL-Based Classifiers:** The adoption of DL classifiers has become a prominent trend in recent literature, owing to their ability to automatically extract hierarchical features and model complex relationships within silent speech data. CNNs, RNNs, and Transformer architectures are frequently employed for this purpose. These DL classifiers excel at recognizing subtle articulatory movements, thereby enhancing the accuracy of silent speech identification. Their applications extend to developing communication aids for individuals with speech disabilities, where the ability to detect minor articulations is crucial for effective voice synthesis and command recognition (Ji et al., 2017).
2. **Sequence-to-Sequence Models:** Sequence-to-sequence models, initially designed for sequence transduction tasks, have recently gained traction as powerful tools for classification in SSR. These models are particularly effective at capturing the temporal dependencies present in silent speech signals, ensuring that the sequential nature of the data is preserved. By maintaining the temporal context, sequence-to-sequence models contribute to improved precision and contextual understanding in silent voice recognition systems, which is especially beneficial in real-time applications, such as silent command interfaces for AR and virtual reality (VR) environments [5].
3. **Ensemble Learning:** Ensemble learning, which involves combining the predictions of multiple models, emerges as a robust classification technique that enhances accuracy and generalization. Recent research explores ensemble methods, including Gradient Boosting and Random Forests, to integrate diverse perspectives and mitigate the limitations of individual classifiers. This approach has proven particularly effective in SSR applications where robustness and high performance are critical, such as in security systems for voice authentication and silent speech-based biometric identification [30].
4. **Transfer Learning:** Transfer learning techniques have gained prominence by enabling models to leverage knowledge from related datasets, particularly those involving audible speech, for silent speech recognition tasks. Pre-trained models, especially those trained on large-scale datasets, are well-suited for SSR, where labeled silent speech data may be scarce. This method significantly improves classification accuracy and accelerates model convergence, making it a valuable tool in applications such as adaptive learning systems for speech therapy, where rapid and accurate model training is essential [47].
5. **Attention Mechanisms:** Attention mechanisms, which allow models to focus on specific segments of silent speech signals, play a crucial role in modern classification techniques. Incorporating attention mechanisms, such as scaled dot-product attention or self-attention, enhances the models' ability to weigh the importance of different spectral or temporal components. This leads to more context-aware and sophisticated classification, which is particularly beneficial in applications like real-time silent command processing in intelligent personal assistants, where the ability to discern relevant features in complex, dynamic environments is critical (Wu et al., [48]).
6. **Contextual Embeddings:** Contextual embeddings have become indispensable in modern classification techniques, providing models with a deeper understanding of the context surrounding silent speech signals. Methods such as BERT (Bidirectional Encoder Representations from Transformers) and ELMo (Embeddings from Language Models) have been adapted for silent speech recognition, offering contextual information that enables classifiers to generate more accurate and nuanced predictions. These embeddings are particularly valuable in conversational AI systems, where understanding the context of silent inputs can significantly enhance the interaction experience, making it more natural and intuitive [20].
7. **Hybrid Classifiers:** Hybrid classification techniques, which combine the strengths of DL and traditional machine learning (ML) methods, are increasingly popular in SSR. These approaches integrate features learned by deep neural networks with those derived from conventional models, such as SVMs or GMMs. This hybridization aims to exploit the interpretability and established performance of traditional models while benefiting from the powerful feature learning capabilities of DL architectures. Such methods are particularly relevant in medical diagnostic tools that rely on silent speech detection, where both precision and interpretability are paramount (Kim et al., [28]).

The advanced classification techniques discussed are finding diverse applications in real-world scenarios, underscoring their significance beyond theoretical research. For instance, DL-based classifiers are increasingly used in developing assistive technologies for individuals with speech disorders, enabling more accurate and responsive communication aids that rely on silent speech recognition. Sequence-to-sequence models, with their ability to capture temporal dynamics, are being deployed in wearable devices for silent command and control, particularly in environments where vocal communication is impractical or impossible, such as in military or space operations. Ensemble learning techniques have found utility in enhancing the robustness of SSR systems used in secure environments, where reliable voice authentication based on silent speech is critical for maintaining security without sacrificing user convenience.

Transfer learning has become indispensable in rapid deployment scenarios, such as creating personalized speech therapy tools that adapt to the unique needs of each user by leveraging pre-trained models. Attention mechanisms and contextual embeddings are driving innovations in smart home systems and AI-driven personal assistants, where the ability to understand silent commands in context-rich environments enhances user experience and system reliability. Lastly, hybrid classifiers are being integrated into diagnostic and rehabilitation tools in healthcare, where they offer a balance of accuracy and interpretability, essential for clinical decision-making and patient care.

In this subsection, we have reviewed advanced methodologies for improving SSR using cutting-edge ML and DL techniques. It highlights the importance of sophisticated feature extraction methods, such as temporal-spectral representations, DL-based representations, and physiological signal integration. These methods enhance the ability to capture intricate patterns and contextual information from silent speech signals. The section also discusses advanced classification techniques, including CNNs, RNNs, Transformer architectures, sequence-to-sequence models, ensemble learning, transfer learning, and attention mechanisms. These classifiers excel at recognizing subtle articulatory movements and improving the overall accuracy and robustness of SSR systems. By combining traditional and DL methods, researchers are capable of developing hybrid approaches that leverage the strengths of both, paving the way for more accurate and effective SSR solutions. The methodologies proposed in this section provide a solid framework for future advancements in silent speech recognition.

9 Comparative performance in the different state-of-the-art works

The performance outcomes of the suggested approaches for SSR are revealed in the results section. Thorough testing demonstrates significant gains in precision, demonstrating the effectiveness of the developed methods. By means of a perceptive examination, these discoveries offer significant perspectives, validating the advancements achieved in raising the bar for silent speech identification. Table 2 presents a thorough evaluation of the effectiveness of the sophisticated feature extraction and classification methods, offering an understanding of the strengths and weaknesses of the developed models.

Conclusively, correlations between discrete language and cognitive metrics from the Comprehensive Aphasia Test and silent speech scores, as determined by group classifications, show significant relationships. These findings provide insights that expand our knowledge of the intricacies present in aphasic situations and advance our understanding of the complex relationship between silent speech and cognitive-linguistic abilities.

10 Limitations and discussion

It is critical to comprehend the status of the field now and project future directions when traversing the intricate terrain of SSR and ML. Instead of ending at a fixed point, the development of SSR is an ongoing process of innovation and refinement. This future-focused perspective is essential for identifying untapped possibilities, innovative ideas, and cutting-edge technologies that may propel SSR ahead into new domains of increased accuracy, increased flexibility, and real-world application. This vision also allows us to proactively solve the limitations and challenges that still exist in the current context, guaranteeing that the progress of SSR in the future will be more than merely revolutionary.

Numerous intriguing directions are indicated by the trajectory of SSR research. Firstly, exploring novel multimodal methodologies combining neuroimaging data, EMG signals, and visual cues may further improve the flexibility and robustness of SSR systems. Moreover, there is promise for comprehending ever-more complex and nuanced silent speech expressions through the integration of contextual information and natural language processing techniques. Investigating real-time apps and enhancing user interfaces is another essential route to take in order to guarantee seamless integration into daily life. Furthermore, cooperative efforts to create standards and benchmarks for

Table 2 Silent Speech Classification Procedures, Evaluation Metrics and Results

Article	Procedure	Evaluation metrics	Result (%)
Ji et al. [24]	The results demonstrate a significant improvement in Word Error Rate (WER) achieved by the Deep Neural Network-Hidden Markov Model (DNN-HMM) strategy compared to the benchmark. The comparison is conducted on the WSJ0 5000-word corpus using 30-dimensional Discrete Cosine Transform (DCT) input features and a decoding strategy consistent with a previous benchmark (SSC benchmark)	WER for HMM WER for GMM	17.4 6.45
[26]	The study evaluates the models using repeated stratified fivefold cross-validation and presents additional metrics such as Cohen's kappa coefficient, precision, recall, and F1-score	Mean accuracy	87
[29]	SpecAugment, a data augmentation technique, is applied to expand the training dataset. The augmented dataset is four times larger, enhancing the generalizability of the model. The use of SpecAugment results in a significant improvement in recognition performance by Character Error Rate (CER) and WER	CER WER	10.1 20.5
[36]	The final SSR system achieved recognition of a 2,200-word vocabulary of more than 1,200 continuous phrases. The phoneme-level recognition models were designed to recognize previously unseen words, showcasing the system's potential for practical applications in consumer and healthcare domains	Recognition rate Error rate	91.1 8.9
[30]	By using additional examples to train the model, the test time performance can be increased. With the exception of vocabulary consisting of ten distinct sentences, the WER increased as the size of the test set vocabulary increased	WER	70–80
[32]	In order to choose the best model that can accurately and precisely predict the spoken phrase and expressed facial expression, the performance of the suggested model is assessed using ML metrics including precision, recall, F1-Score, and support	Precision Recall F1 score Support	89 97 93 61
[23]	Sensor is located under the chin and in front of the lips	HMM Accuracy	59.0
[6]	The system was built on a subject-dependent architecture, meaning that each participant's classifier was trained using training data specific to them. They used the Adam optimizer to train each of the three neural networks	Accuracy	88–92
[44]	They used each patient's three utterances as a single training sample, for a total of 84 training samples. Determining if a particular speaker had MCI or not was the training goal, leading to a 2-class classification task	Accuracy Precision Recall Specificity F1 AUC	75.0 76.5 81.3 66.7 78.8 67.6%
[21]	The Deep Speech system was contrasted with a number of commercial speech systems, including wit.ai, Google Speech API, Bing Speech, and Apple Dictation The purpose of the test is to benchmark noise-induced performance	WER	11.85

assessment metrics would encourage a more uniform and similar assessment of SSR models between research.

Some obstacles still exist even with advancements toward increased efficacy and application. SSR models are limited in their potential to be broadly applicable since it can be challenging to gather large and diverse datasets, particularly for certain user groups. Enhancing model transparency and trust is crucial, as there remains a worry over the interpretability of DL architectures. The ethical concerns of permission and privacy in the gathering and use of silent speech data also require ongoing consideration. In order to steer SSR research toward a future characterized by inclusiveness, reliability, and responsible deployment in a variety of real-world contexts, it will be imperative to address these constraints.

Although the feature extraction and classification methods examined in this work offer fundamental understandings, their application in many contexts reveals some drawbacks. The optimization approaches' performance is contingent on the environment and may differ for various datasets and problem areas. These drawbacks highlight the need for the suggested methods to be further improved, especially in incorporating more flexible and broad models. Furthermore, the methods of categorization, while useful in some situations, would need to be improved in terms of robustness and scalability in order to be applied more widely. We recognize these difficulties and suggest that going forward, efforts should concentrate on creating more adaptable strategies that deal with these pointed restrictions. Thus, the foundation for these developments is laid by this study.

11 Summary

Examining SSR in the context of ML reveals a dynamic story that traces the development from early waveform analysis to current robust ML methods. Even while SSR has the potential to completely change communication paradigms, there are still a number of unmet research needs and obstacles. Future research should focus on improving the SSR systems' resilience and flexibility, addressing data usage ethics, and investigating real-time, continuous SSR applications. Proactive measures are required going ahead to solve these research obstacles and shortcomings. Utilizing developments in ML, signal processing, and cognitive sciences, collaborative multidisciplinary research is crucial. In future, research should concentrate on creating unique approaches for multimodal data fusion, improving SSR system interpretability, and investigating cutting-edge uses in human–computer interaction, healthcare, and education. To sum up, this work acts as a compass to direct the course of SSR research in the future. In order to promote more inclusive and efficient communication solutions, we hope to stimulate more developments in SSR technology by pinpointing the primary research gap and suggesting specific research avenues. SSR has a bright future ahead of them as they continue to push the limits of ML and human–machine interaction.

12 Recommendations

The study of current ML techniques for silent speech recognition reveals several suggestions in this area. In order to improve accessibility for people with speech problems, it is critical to investigate non-invasive technologies like wearables and ultrasonography while developing silent speech interfaces. To ensure that these technologies are effective for meeting healthcare needs—particularly in illnesses like multiple sclerosis—clinical environments for validation and adaptation are essential. It is imperative that EMG-based interfaces continue to be prioritized, with a focus on their affordability, non-invasiveness, and suitability for treating speech impairments. Healthcare-specific silent speech interfaces, such as AlterEgo, can be customized to help patients with communication difficulties, especially those who do not move much when they speak silently. Promoting the thorough validation of smartphone-based interfaces, such as "EchoWhisper," significantly advances the industry by ensuring that they outperform current technologies in a variety of real-world circumstances.

Author contributions Adiba Tabassum Chowdhury: Conceptualization, Methodology, Software, Writing – original draft, Writing – review & editing. Mehrin Newaz: Methodology, Software, Writing – original draft, Writing – review & editing. Purnata Saha: Methodology, Software, Writing – original draft, Writing – review & editing. Mohannad Natheef AbuHaweeleh: Methodology, Software, Writing – review & editing. Sara Mohsen: Conceptualization, Methodology, Software, Validation. Diala Bushnaq: Conceptualization, Writing – original draft, Writing – review & editing. Malek Chabbouh: Writing – original draft, Writing – review & editing. Raghad Aljindi: Conceptualization, Methodology, Software, Validation. Shona Pedersen: Validation, Methodology, Software, Writing – review & editing. Muhammad E.H. Chowdhury: Conceptualization, Methodology, Supervision, Writing – original draft, Writing – review & editing.

Funding Open Access funding provided by the Qatar National Library. This work was made possible by Undergraduate Research Experience Program grant # UREP29-043-3-012 from Qatar National Research Fund (QNRF) and is also supported by Qatar University Student Grant: QUST-1-CENG-2024-1722. The statements made herein are solely the responsibility of the authors. The open-access publication cost is covered by Qatar National Library.

Data availability Data sharing is not applicable to this article as no datasets were generated or analyzed during the current study.

Declarations

Conflicts of interest Authors have no conflict of interest to declare.

Institutional review board Not applicable.

Informed consent Not applicable.

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